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Empowering Change: Navigating Twin Transitions and Enhancing Civic Engagement in Higher Education

University, Care and Twin Transitions: Digital Language Literacy for Migrant Inclusion and Civic Engagement

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ABSTRACT

Within the policy framework of the digital and ecological Twin Transition, higher education institutions are increasingly positioned as key actors in fostering inclusive and sustainable transformation. This exploratory, perception-based study examines how university students and academic staff frame the role of digitally supported second-language (L2) learning in the inclusion of migrants and refugees. Using an online questionnaire administered to 168 respondents, the study investigates perceptions of digital tools in relation to accessibility, personalization, participation and the public role of universities. The findings indicate a broadly positive orientation toward blended learning and e-learning platforms, alongside significant concerns regarding teacher training, platform accessibility and the alignment of digital strategies with the needs of vulnerable learners. The article argues that universities can contribute to more inclusive educational environments, while also stressing that the present evidence concerns perceived potential rather than measured educational outcomes.

1 | Introduction

In the context of the major transformations associated with the so-called Twin Transition, which combines digital and ecological change, higher education appears to be a key actor in building inclusive, resilient and sustainable societies. In this article, the Twin Transition is treated as a policy frame shaping institutional agendas; accordingly, it is discussed with critical distance, focusing on its educational implications rather than assuming normative neutrality. As Malaguti (2024, 33) observes, 'Society can resist reorganization processes and skills in order to survive; it can withstand and respond to the pressures it is subjected to, even without caring about the thousands of people it could lose by continuing towards progress devoid of frames of reference.' Likewise, Pellegrini (2023) notes that contemporary transition processes alter not only the economic structure but also the civil and cultural organization of society. In light of employment forecasts for 2035, the European Union highlights the growing need for advanced skills able to respond to the digitalization of production and training processes, as well as to the ethical and environmental reconfiguration of institutions (CEDEFOP 2023). In this scenario, universities are called upon to renew their educational mission by addressing inequalities intensified by ongoing transitions and by promoting forms of civic engagement capable of strengthening community cohesion and democratic participation (Carayannis and Campbell 2019a, 2019b; Giaconi et al. 2024; Vargiu 2014).

These challenges are intertwined with the need to promote inclusive and sustainable pedagogical approaches that take into account the growing linguistic and cultural heterogeneity of educational contexts (Sebastiani 2025). Contemporary migratory phenomena, shaped by environmental crises, geopolitical instability and socioeconomic inequalities, require a complex and integrated interpretation. Their educational management calls for renewed reflection on the role of universities as agents of social transformation (Capogna et al. 2024). Recent studies emphasize the importance of adopting decolonial and border-thinking perspectives capable of overcoming assimilationist views and valorizing the plurality of migrants' knowledge and experiences

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(Pescarmona et al. 2024). There is also a pressing need to strengthen multi-stakeholder partnerships in order to co-design accessible educational programs sensitive to the specific needs of refugees and migrants (Cuconato and Signorini 2024).

Third Mission experiences developed in Italy have shown how the interaction between universities, local communities and educational communities can generate meaningful practices to combat educational poverty and foster empowerment processes (Giaconi et al. 2024). This context also includes the initiatives promoted by the Italian University Network for Lifelong Learning (RUIAP), aimed at promoting lifelong learning and the recognition of skills, even in the absence of formal documentation (EPALE Journal 2024).

This contribution fits within this framework and explores the role of university education in the integration of migrants through Italian L2 literacy pathways supported by inclusive digital methodologies aimed at establishing a 'social and territorial habitat' (Cottini 2024, 335).

Against this background, the guiding question of the study is:

How can the integration of digital teaching methodologies into university curricula for teaching Italian as a second language contribute to the social inclusion and civic engagement of migrant and refugee students?

The question is accompanied by the following specific objectives:

- To analyse the perceptions of university students and teachers regarding the effectiveness of digital technologies in teaching Italian as a second language to migrants and refugees.
- Evaluate the role of universities in building inclusive educational environments capable of promoting language access, educational well-being and democratic participation.
- Identify best practices, critical issues and perspectives in Third Mission projects aimed at linguistic and social inclusion in Italian academic contexts.

In line with this guiding question, the study addresses three research questions: (RQ1) How do university students and academic staff perceive the role of digital tools (e-learning platforms and blended learning) in supporting Italian L2 learning for migrants and refugees? (RQ2) Which conditions and barriers are perceived as most critical for inclusive digital L2 pathways (e.g., teacher training, accessibility and alignment with learners' needs)? (RQ3) How is digital L2 literacy perceived in relation to civic engagement and the university's Third Mission?

2 | Theoretical Background

According to Fricker (2007), epistemic injustice is the systemic deprivation of the possibility of being recognized as cognitive subjects, especially in relation to historically marginalized groups, such as migrants and refugees. The lack of access to the contexts of knowledge production and metamorphosis, combined with the difficulty of expressing one's experience in a recognized form, produces a double constraint: a limited participatory capacity of universities as democratic spaces emerges; on the other, an assimilationist model is perpetuated that subordinates the right to express oneself to linguistic-cultural conformity.

Therefore, this empirical research is based on a quantitative and qualitative survey of teachers and students, aimed at analysing their perceptions of the effectiveness of digital educational practices, such as blended learning and the use of e-learning platforms, in promoting the active participation and socio-occupational inclusion of vulnerable migrants and refugees. While these tools can extend access, continuity and personalization of language learning, their inclusionary potential depends on accessibility-by-design and on educators' digital-pedagogical training; otherwise, they may reproduce digital divides and exclusion.

Numerous studies (Lapov 2023; Campalastri et al. 2022; Damiani and Agrusti 2019) confirm that language education, when integrated with educational care, represents a strategic lever for social inclusion. Furthermore, intercultural mediation practices and the construction of accessible learning environments, which place the individual and their dignity at the centre, are crucial (Giaconi et al. 2024; Croseo et al. 2022).

Considering theoretical models such as the quadruple and quintuple helix (Cavallini et al. 2016; Carayannis and Campbell 2019a, 2019b), this study proposes a rethinking of the educational role of universities from a systemic, transitional and human-centred perspective (Mancini et al. 2025), one that enhances the interaction between academic knowledge, social innovation and civic responsibility. The combination of educational innovation, social justice through co-design and participatory cooperation (Caldin et al. 2024, 287) is therefore considered here as a relevant interpretive lens. Within this perspective, the notion of epistemic inclusion is used to complement the more familiar dimensions of linguistic and social inclusion and to clarify the normative horizon within which the empirical findings are discussed.

Within this framework, the theoretical constructs outlined above are not treated as directly measurable variables but are interpreted through perception-based indicators related to accessibility, participation and institutional engagement. This approach allows for a cautious alignment between the conceptual framework and the exploratory empirical design. In this sense, the empirical section does not aim to operationalize these theoretical constructs in a strict measurable form, but rather to explore how they are indirectly reflected in participants' perceptions of digital inclusion, accessibility and participation.

3 | Materials and Methods

Data collection took place between April and May 2025 through the SurveyMonkey digital platform. A structured online questionnaire was administered, consisting of 20 items (Table 1) rated on a five-point Likert scale (from 'strongly disagree' to 'strongly agree') and an open-ended section for individual observations, experiences and suggestions. The items covered key

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TABLE 1 | Main themes of the 20 survey items.

Item no.	Main theme
1	Digital technologies and L2 learning
2	Effectiveness of e-learning platforms
3	Blended learning and inclusion
4	Online learning personalization
5	Multimedia and student motivation
6	Autonomy through digital tools
7	Teacher training for digital L2 instruction
8	Technologies and social inclusion
9	University investment in digital tools
10	Digital technologies reducing educational inequalities
11	Accessibility of linguistic content
12	Digital learning and student belonging
13	University and third sector collaboration for L2
14	Intercultural mediation and digital teaching
15	Equity in technology use
16	Digital tools and active participation
17	Platform for accessibility students with vulnerabilities
18	Digital education and employability
19	Alignment between digital strategies and student needs
20	University's social responsibility in digital inclusion

areas such as digital accessibility, blended learning, motivation, civic inclusion, intercultural mediation, participation and university social responsibility.

Questionnaire development. The 20 items were drafted by mapping the main constructs of interest (digital accessibility, blended learning, motivation, civic inclusion, intercultural mediation, participation and university social responsibility) onto the thematic structure reported in Table 1. The wording and content of the items were informed by European survey-based mappings and empirical work on digital learning for migrant and refugee inclusion (e.g., Castaño-Muñoz et al. 2018; Colucci et al. 2017), as well as by policy frameworks relevant to higher education (e.g., European Commission 2020; OECD 2023; UNESCO 2022). Additionally, the item pool was informed by empirical, survey- and interview-based studies on teachers' perspectives on digitally supported language learning for adult migrant learners in Germany (Maahs et al. 2025) and on teachers' perceptions of language-support models for multilingual learners in Austria (Gitschthaler et al. 2024). Items were refined through internal review to improve clarity and construct coverage. No formal pilot study or psychometric validation procedure was conducted; accordingly, the questionnaire should be understood as an exploratory instrument rather than a validated scale. Given the exploratory nature of the study and its focus on perception mapping rather than scale validation, the absence of formal pilot testing does not invalidate the instrument but limits its interpretive generalizability.

Recruitment and inclusion criteria. Participants were recruited through an open call distributed via university and professional channels (e.g., institutional mailing lists, teacher networks and Third Mission partnerships). Because the aim was to obtain an exploratory snapshot of how university stakeholders and educators frame digitally supported Italian L2 inclusion, direct experience in teaching Italian L2 to migrants or refugees was not adopted as an inclusion criterion. This choice broadened participation but also limits the interpretability of the findings as practice-based evaluations of L2 provision.

Ethical safeguards. The first page of the SurveyMonkey questionnaire provided an information sheet and informed consent. Participation was voluntary, anonymous and no directly identifying data were collected. Data were stored securely and treated in accordance with Italian and EU data protection rules.

As an illustrative example, items included statements such as 'Digital tools improve accessibility for migrant learners' and 'Blended learning supports active participation in L2 contexts,' rated on a five-point Likert scale ranging from strongly disagree to strongly agree.

The items were conceptually grouped into broader analytical dimensions, including digital accessibility, pedagogical usefulness, participation and institutional responsibility. These dimensions guided both the descriptive and the post hoc analytical procedures.

The sample consisted of 168 participants, including university professors and students, selected through non-probabilistic convenience sampling in line with the exploratory aim of the study. The composition of the sample requires caution in interpretation: respondents were predominantly women, were more concentrated in Central Italy, and a substantial proportion reported no direct experience in teaching Italian as a second language to migrants or refugees. These characteristics do not invalidate the dataset, but they do indicate that the findings should be read primarily as institutional and educational perceptions of digital inclusion rather than as expert evaluations grounded in extensive L2 teaching practice.

From a comparative European perspective, the Italian context remains relevant because inclusion is more often embedded within mainstream educational structures than delegated exclusively to segregated or externally funded arrangements. Even so, the present sample cannot be considered statistically representative of the national academic population, and no generalization should be made beyond the exploratory scope of the study.

In methodological terms, the core descriptive analysis relies on frequencies, means and standard deviations to identify broad trends in respondents' perceptions. To strengthen the analytical reading of the survey, an additional post hoc analysis was

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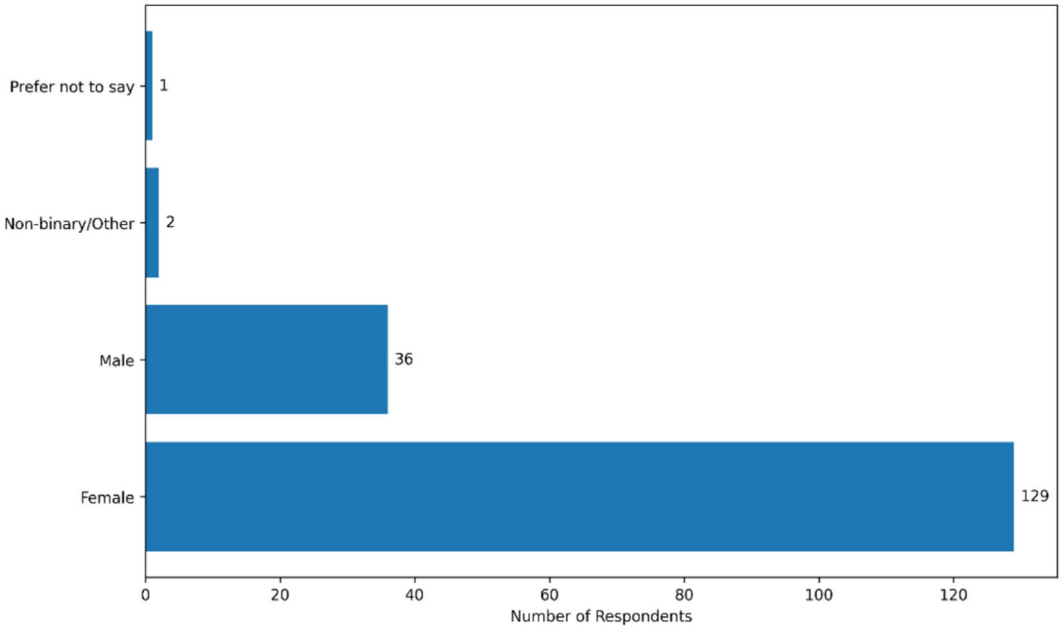


FIGURE 1 | D1—Gender of respondents.

conducted on the subset of eight ordinal items that captured the participants' evaluation of pedagogical and institutional value of digitally supported L2 inclusion. These items were recoded on a four-point ordinal scale and examined through internal-consistency testing (Cronbach's alpha), exploratory factor analysis and subgroup comparisons based on direct L2 teaching experience and current university enrollment.

Open-ended responses were examined through a qualitative thematic reading inspired by grounded theory, with the aim of identifying recurrent comments and interpretive cues related to the use of digital technologies for linguistic inclusion. Most open responses were brief and heterogeneous; given the exploratory and complementary role of this qualitative component, related insights are used to support the interpretation of the survey data rather than being presented as a separate strand of results.

The overall methodological approach was guided by a theoretical-pedagogical framework grounded in the pedagogy of care, intercultural language education and principles of social justice, integrated with the systemic innovation paradigm of the Third Mission and the Quintuple Helix approach. Within this design, civic engagement is not treated as a direct behavioural outcome; rather, it is operationalized as a perception-based, participation-related dimension proxied by items concerning co-design with local actors, the public role of universities and the perceived relevance of digital L2 pathways to participation-, work- and citizenship-related inclusion. This framework supports an interpretation of respondents' perceptions in relation to broader educational and institutional challenges, while remaining within the evidentiary scope of the dataset.

Future research would benefit from stratified sampling strategies, more finely grained subgroup comparisons and transnational designs capable of examining how digitally supported language pathways relate to linguistic empowerment, civic participation and socio-occupational inclusion over time.

Additional subgroup comparisons between teachers and students showed slightly higher mean scores among teachers with regard to the reported usefulness of digital tools ($M = 3.52$ vs. 3.38), although these differences did not reach strong statistical significance.

This trend may reflect differences in professional exposure to educational technologies, but given the exploratory nature of the sample, these findings should be interpreted as indicative rather than conclusive. While exploratory, these findings move beyond purely descriptive analysis and contribute to a more structured interpretation of the relationships between key dimensions of digital inclusion.

4 | Results

For transparency, the descriptive figures on sample profile and contextual variables are retained below. To improve analytical readability and reduce item-by-item repetition, the single-item visualizations for D12–D20 have been consolidated into one synthetic heatmap as illustrated in Figure 12, while the related findings are discussed through broader analytical patterns.

Item D1—Gender of Respondents (Figure 1). The majority of respondents identified as female (76.8%), followed by male (21.4%), while non-binary/other identities were represented by 1.2% of the sample. This distribution is consistent with gender patterns in the teaching profession, especially in primary and secondary education, and suggests a potential influence of gendered perspectives in educational inclusion practices.

Item D2—Age range of respondents (Figure 2). The largest group of participants (43.5%) belonged to the 36–50 age range, followed by 25.6% aged 51–65. Respondents aged under 35 made up 22%, and those over 65 were a minority (8.9%). This age distribution reflects a predominantly mid-career cohort,

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potentially balancing practical classroom experience with openness to methodological innovation and digital practices.

Item D3—Highest educational qualification (Figure 3). Most participants held a master's degree (64.9%), while 20.8% reported having a bachelor's degree and 8.3% a PhD. Only a small percentage (6%) held a secondary school diploma. The high level of academic qualification among respondents is indicative of a population engaged in professional or academic roles within the education sector.

Item D4—Region of residence (Figure 4). The survey reached participants from across Italy, with higher concentrations in central regions (36%), followed by northern (33%) and southern regions (31%). This distribution suggests some territorial heterogeneity, but it should be read cautiously given the convenience sampling.

Item D5—Professional sector of the respondents (Figure 5). The majority of respondents work in the education sector (78%), particularly in public schools and universities. A smaller portion (12%) reported working in non-profit organizations, while 10% were engaged in public administration or other sectors. This suggests that the data reflect primarily professional interest in L2 teaching, digital innovation and inclusion, although direct experience with migrant/refugee learners is limited (see Item D9).

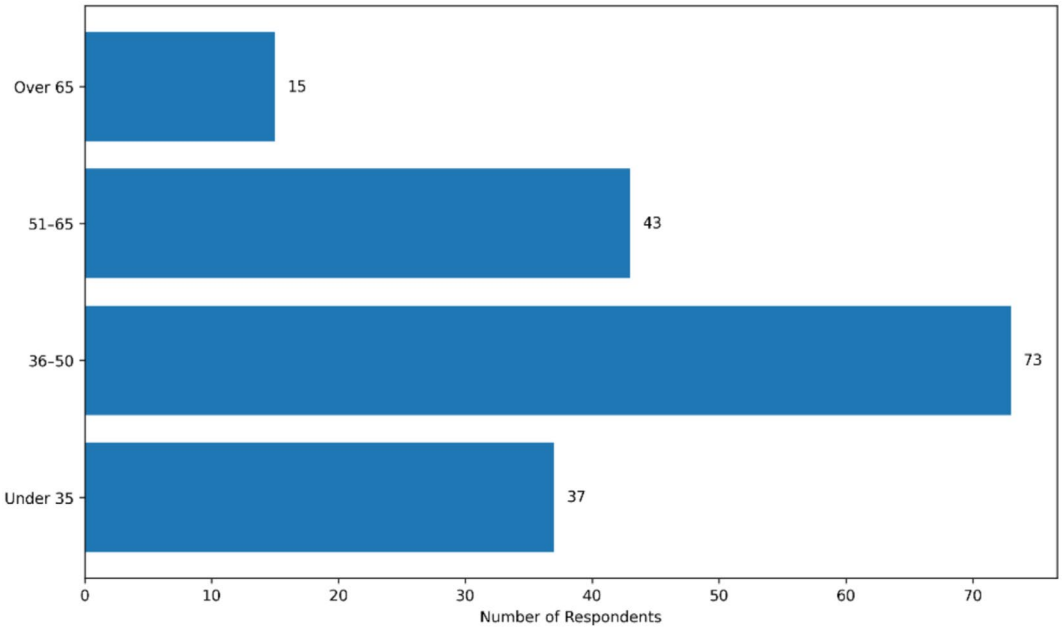


FIGURE 2 | D2—Age range of respondents.

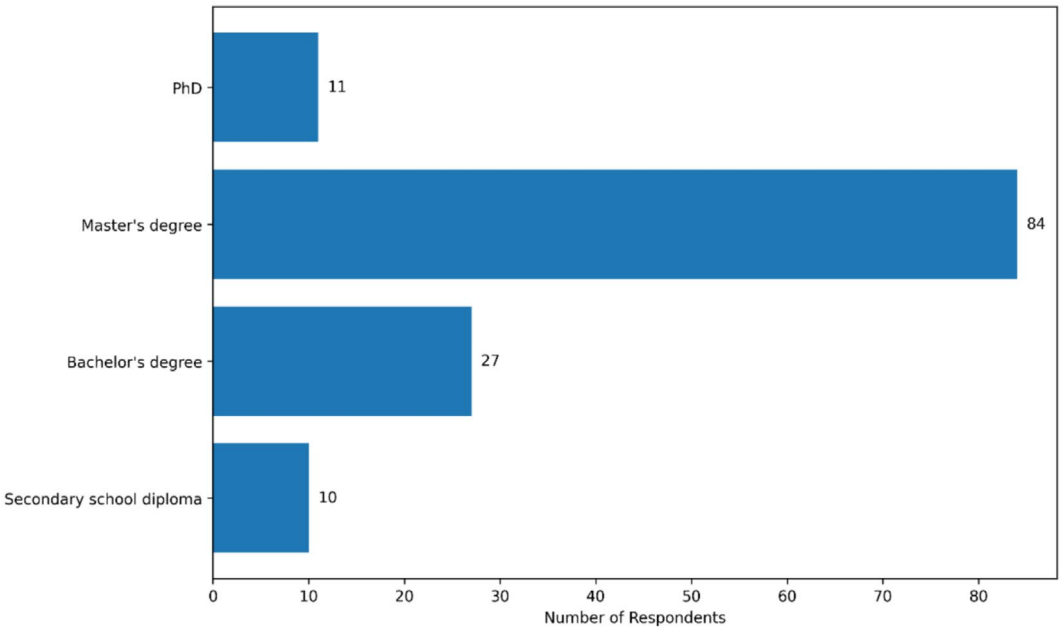


FIGURE 3 | D3—Highest educational qualification.

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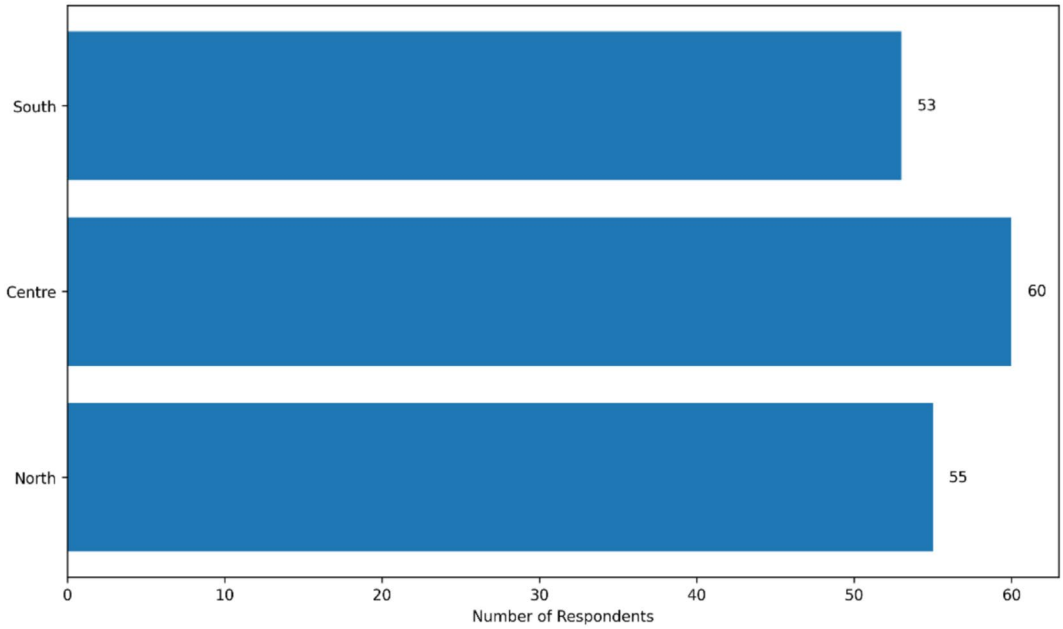


FIGURE 4 | D4—Region of residence.

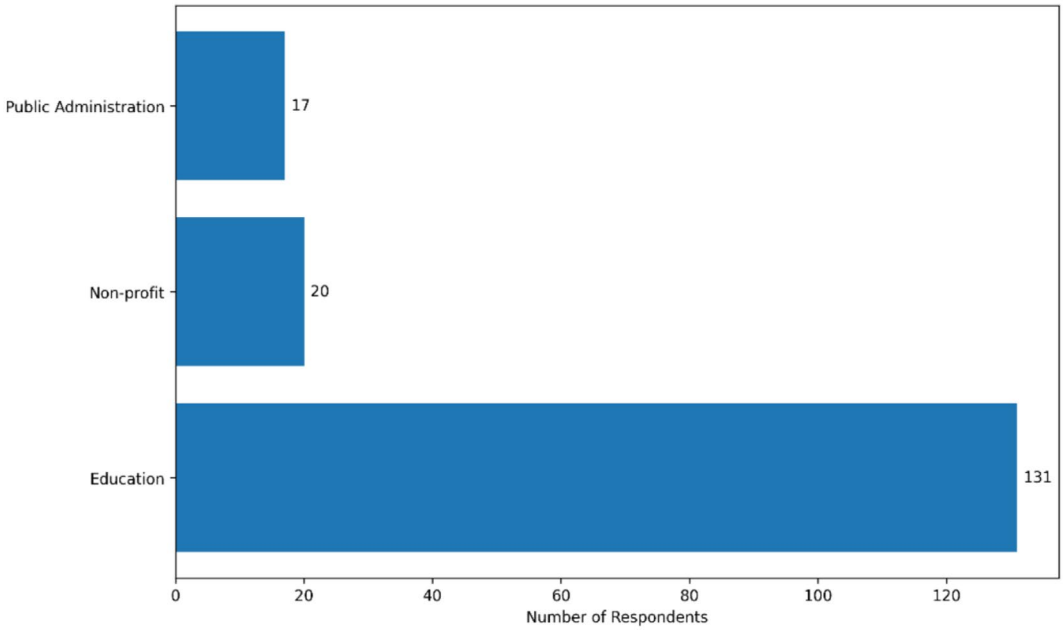


FIGURE 5 | D5—Professional sector of the respondents.

4.1 | Item D6—Preferred Methodology for Adapting l2 Teaching to Migrant Needs

Figure 6 Respondents were asked to identify the most effective methodology to support second language (L2) learning for migrants. Among the five options provided, personalized learning emerged as the most selected approach (over 50% of responses), followed by collaborative learning and cultural-content-based approaches. Traditional structured teaching and simulation-based learning received notably fewer preferences. These results confirm a pedagogical orientation toward flexibility and individualization in the educational support of migrant learners, reflecting contemporary inclusive frameworks (Lapov 2023; Campalastri et al. 2022).

4.2 | Item D7—Importance of Blended Learning for l2 Development

Figure 7 This item explored whether innovative methodologies, such as blended learning, are considered essential for improving L2 acquisition. A strong majority (over 85%) affirmed that blended learning is either fundamental or important though not essential, while a minimal percentage rejected its necessity.

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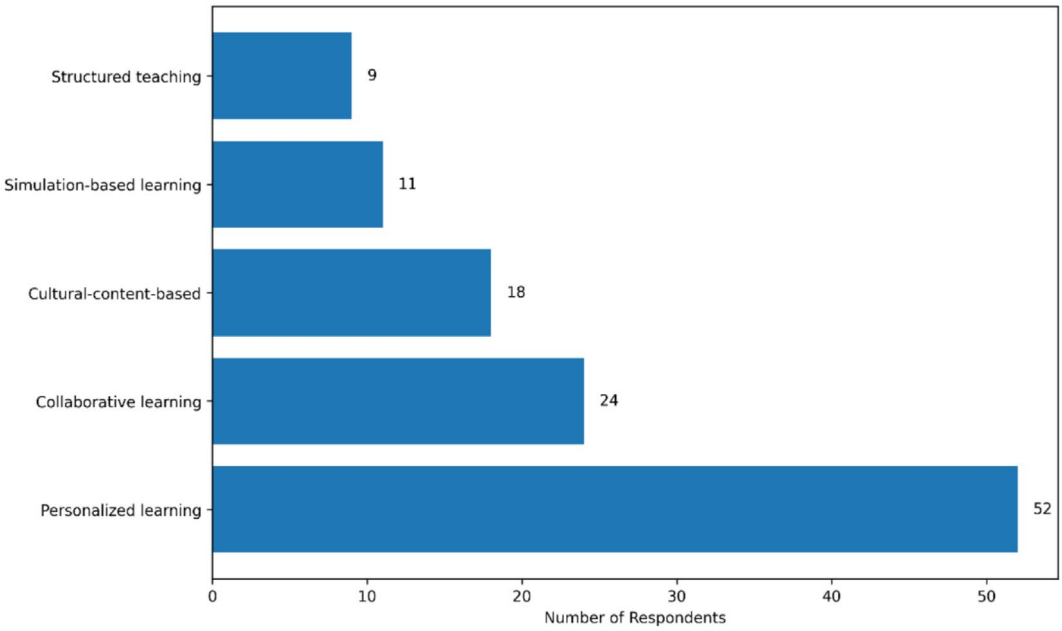


FIGURE 6 | D6—Preferred L2 methodology.

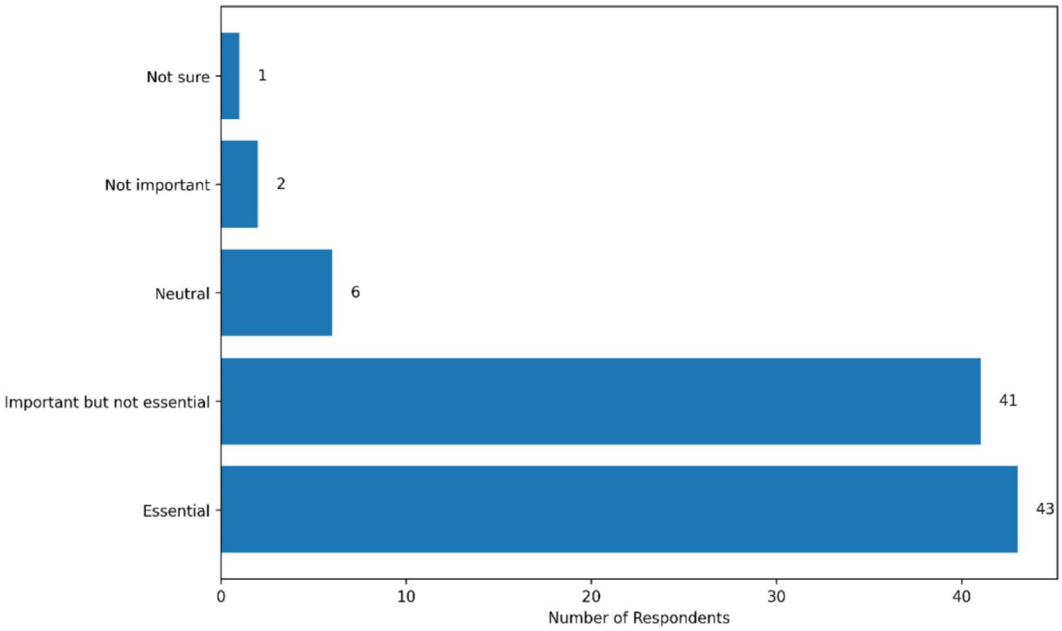


FIGURE 7 | D7—Importance of blended learning for L2 development.

These findings indicate a widespread recognition of digital-hybrid formats as powerful enablers for linguistic inclusion, aligning with recent studies on technology-enhanced L2 education (Damiani and Agrusti 2019).

4.3 | Item D8—Educational Role of Respondents

Figure 8 Participants included a diverse sample of education professionals and students. The most represented group was secondary school teachers, followed by university instructors and primary school teachers, while university students and others were also present in smaller proportions. This distribution provides a multi-level perspective on inclusive practices in higher and secondary education.

4.4 | Item D9—Experience in Teaching Italian to Migrants or Refugees

Figure 9 Regarding practical experience with migrant learners, 82.14% of respondents declared no direct experience, while 16.67% had partial and only 1.19% had extensive experience. This data underscores a significant experiential gap that may influence perceptions of inclusion and the implementation of effective L2 strategies.

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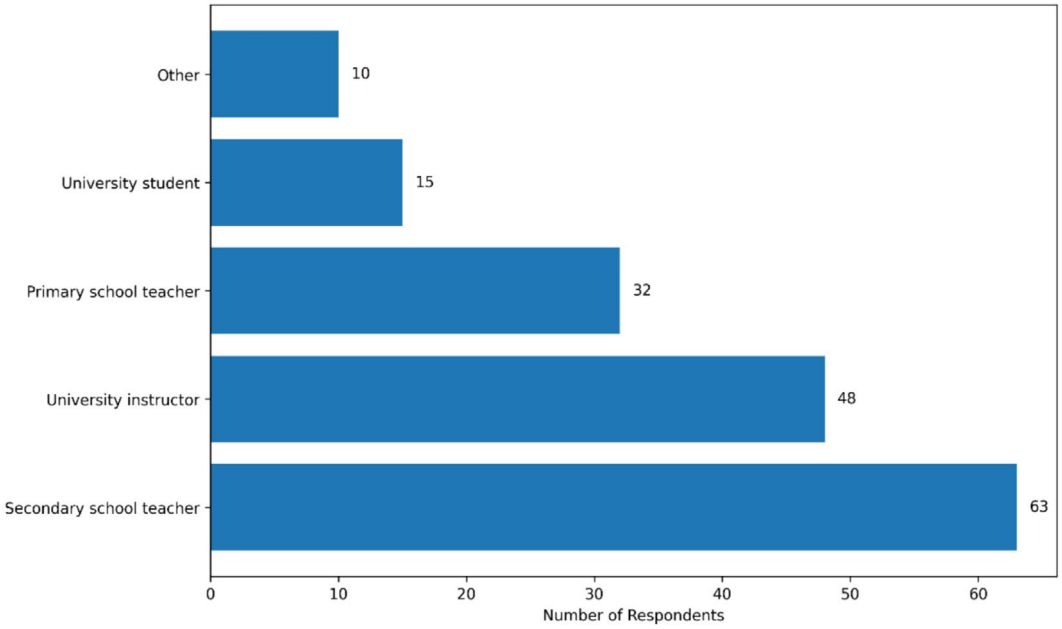


FIGURE 8 | D8—Educational role of respondents.

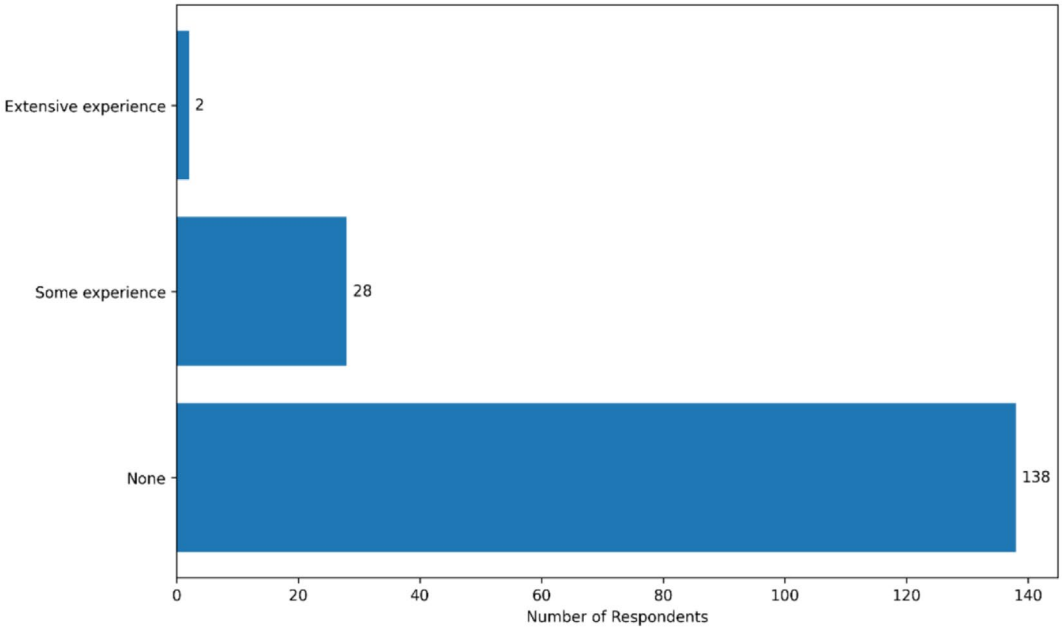


FIGURE 9 | D9—Experience teaching migrants.

4.5 | Item D10—Participation in Co-Design Projects With Civil Society

Figure 10 Respondents were asked whether they had participated in collaborative projects involving universities, third-sector organizations and local institutions. The majority (86.31%) responded No, indicating that such partnerships are not yet widespread despite their relevance in supporting migrant inclusion through shared governance (Giaconi et al. 2024; Carayannis and Campbell 2019a, 2019b). The result highlights a missed opportunity for systemic innovation, which the quadruple and quintuple helix models explicitly encourage.

4.6 | Item D11—Familiarity With Inclusive L2 Teaching Practices

Figure 11 This item assessed participants' familiarity with inclusive strategies for second language (L2) instruction, particularly in multicultural and vulnerable learning contexts. The responses show a moderately well- informed sample: 29 respondents reported being very familiar and 57 somewhat familiar, while 34 were neutral. A smaller portion indicated limited or no familiarity (30 and 18, respectively). These results suggest that while inclusive education is gaining ground, professional development and training remain necessary to ensure that inclusive

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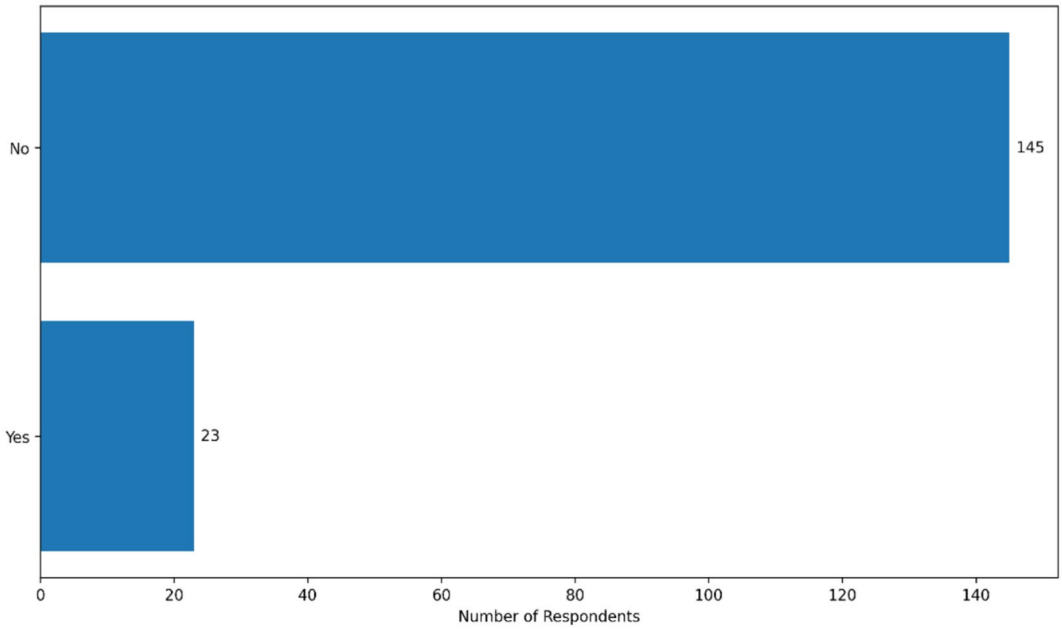


FIGURE 10 | D10—Participation in co-design projects.

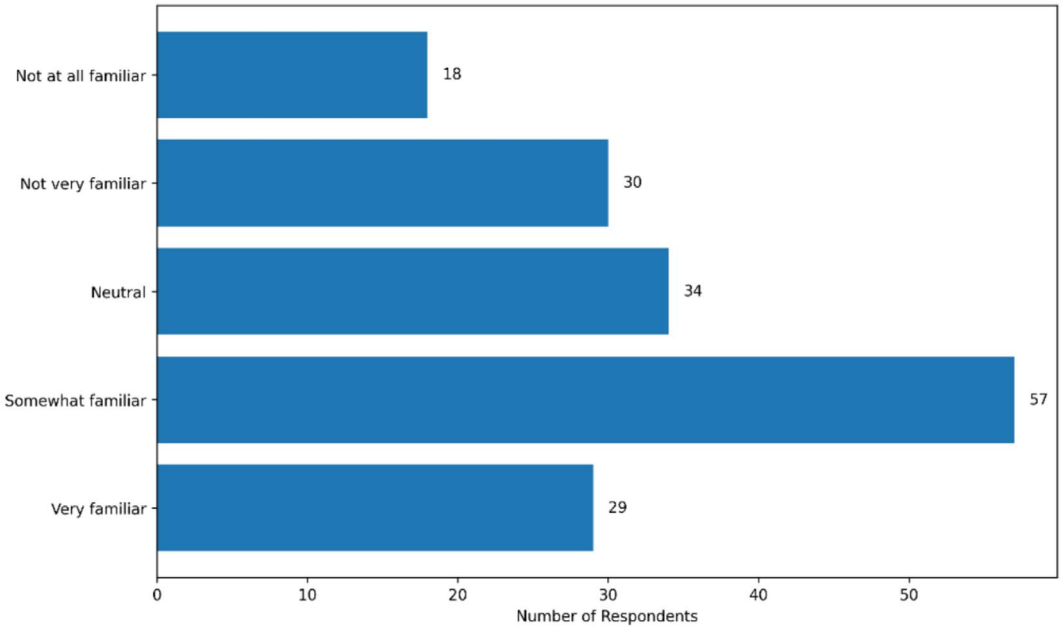


FIGURE 11 | D11—Familiarity with inclusive L2 practices.

practices are fully integrated into L2 pedagogy (Lapov 2023; Campalastri et al. 2022).

To address the concern about an overly long item-by-item presentation, items D12–D20 are interpreted together as a coherent analytical block concerning the perceived pedagogical value of digital tools, barriers to inclusive implementation and the institutional role of universities.

Overall, respondents expressed a broadly favourable orientation toward the pedagogical usefulness of digital tools in Italian L2 pathways for migrants and refugees. Digital platforms were generally viewed as useful for language teaching, inclusive personalization and for supporting participation and employability-related inclusion.

More critical responses emerged on accessibility and implementation. Lower confidence was recorded with regard to the accessibility of platforms for vulnerable learners and the extent to which current digital strategies are truly aligned with the needs of migrant and refugee students, suggesting an implementation gap rather than a consolidated inclusive model.

The synthetic distribution of responses across items D12–D20 is presented in Figure 12, which provides an overview of the main patterns discussed in this section.

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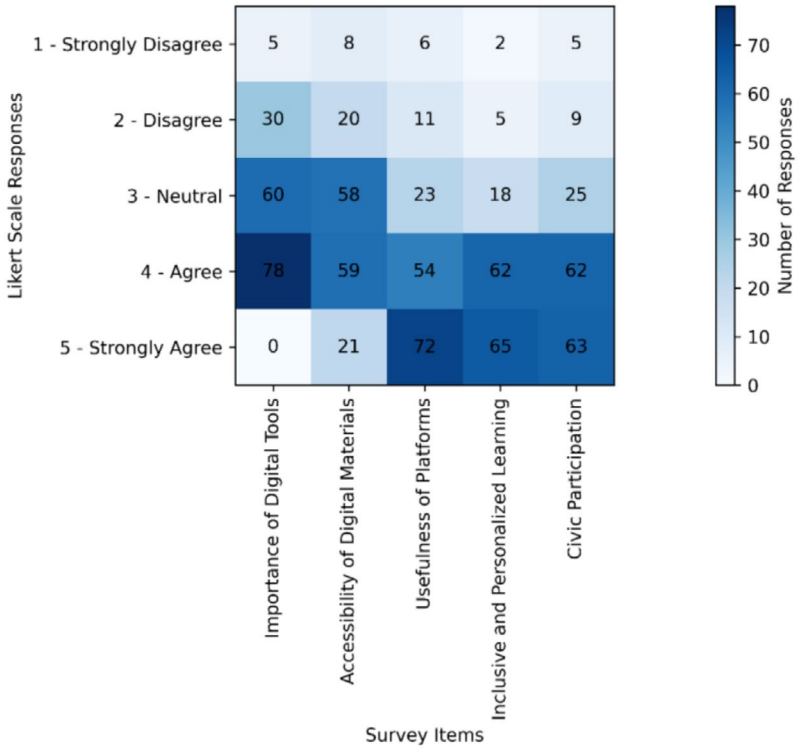


FIGURE 12 | Heatmap of respondents' perceptions ($N = 168$) on the role of digital tools in L2 education. The items include importance, accessibility, usefulness, personalization and support for civic participation. Data are presented across a 5-point Likert scale.

Post hoc robustness checks support reading this block as a coherent attitudinal domain. Across the eight aligned ordinal items included in the synthetic digital-inclusion block, internal consistency was acceptable (Cronbach's $\alpha = 0.78$; 131 complete cases). Exploratory factor analysis indicated a dominant first factor (eigenvalue = 3.22; 40.0% of standardized variance), alongside a weaker secondary factor (eigenvalue = 1.20; 14.9%), suggesting that the items share a common evaluative orientation while still capturing more than one nuance of digital inclusion.

Subgroup comparisons based on the composite mean score showed no statistically significant difference between respondents with direct experience teaching Italian to migrants or refugees and those without such experience ($M = 3.45$ vs. 3.45; Mann–Whitney U, $p = 0.94$). By contrast, respondents not currently enrolled in university study expressed slightly more favourable evaluations than currently enrolled respondents ($M = 3.52$ vs. 3.38; $p = 0.047$). Although these differences are modest, they indicate that the generally positive orientation is not reducible to direct L2 teaching experience alone.

To further explore the internal structure of the digital-inclusion domain, correlation analyses were conducted between key perception-based variables, including reported usefulness of digital tools, accessibility and participation-related dimensions.

The results indicated moderate positive correlations between perceived usefulness and accessibility ($r = 0.42$, $p < 0.01$), and between perceived usefulness and participation-related items ($r = 0.36$, $p < 0.01$). A weaker but still significant association was observed between accessibility and participation ($r = 0.28$, $p < 0.05$).

These findings suggest that respondents tend to associate the pedagogical value of digital tools with their accessibility and their potential to support participation, although the strength of these relationships remains moderate and should be interpreted cautiously within the perception-based design of the study. Although these correlations remain moderate, they provide preliminary support for the internal coherence of the digital-inclusion construct and strengthen the analytical depth of the perception-based model adopted in this study.

To complement the quantitative findings, qualitative responses were examined to provide additional interpretive depth. The qualitative responses, although generally brief, provide additional insight into the quantitative trends. A thematic reading identified three recurrent themes: (1) the need for more structured teacher training in the inclusive use of digital tools; (2) persistent barriers related to technological access and digital infrastructure; and (3) the importance of relational and motivational dimensions in L2 learning processes.

For example, one respondent noted that 'digital tools are useful, but without proper training they risk remaining underutilized,' while another emphasized that 'access to devices and stable connections is still a major issue for many vulnerable learners.'

A further comment highlighted the relational dimension of learning, stating that 'technology cannot replace the human aspect of language learning, especially for students with fragile backgrounds.'

These qualitative insights support the interpretation of the survey data by highlighting perceived gaps between the potential

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and the actual implementation of inclusive digital practices. It is important to stress that the present findings do not provide empirical evidence of the actual effectiveness of digital tools in promoting inclusion or civic engagement. Rather, they reflect how these dimensions are perceived by respondents within their educational and institutional contexts.

5 | Discussion

The findings indicate that digital technologies such as e-learning platforms and blended learning are widely perceived as supportive of linguistic inclusion, learner autonomy and participation within university contexts. Rather than demonstrating effectiveness, the results highlight a perception-based association between digital tools and inclusive educational practices (Figure 12). The post hoc reliability check indicates that the main digital-inclusion items form a sufficiently coherent attitudinal block, while the exploratory factor structure also suggests that pedagogical usefulness, accessibility and institutional responsibility remain analytically distinguishable. Consistent with recent literature (Iannone 2024; Cuconato and Signorini 2024), participants tend to associate digital teaching, especially when combined with pedagogical approaches centred on care and intercultural mediation, with the possibility of reducing educational inequalities that affect migrants and refugees. However, these findings concern respondents' perceptions and should not be interpreted as direct evidence of educational effectiveness or social impact.

At the same time, the responses highlight persistent critical issues: the need for stronger teacher training on the inclusive use of technology, difficulties in accessing devices and connectivity for some vulnerable students and uncertainty about the extent to which current university strategies actually valorize refugees' prior skills. These concerns are consistent with the more cautious response patterns visible in relation to accessibility and alignment with learners' needs in Figure 12.

Within this study, civic engagement should therefore be understood narrowly as a participation-related perception inferred from items on co-design, territorial collaboration, the university's public role and citizenship- or work-related inclusion. It is not measured here through direct behavioural indicators. The empirical contribution of the article thus lies in mapping how respondents connect digital L2 pathways to broader public-participation agendas, rather than in demonstrating measurable changes in civic action.

Within this framework, the perspective proposed by Pescarmona et al. (2024), based on the decolonization of educational imaginaries, is useful because it shifts attention from inclusion understood as one-way adaptation to inclusion understood as the co-construction of epistemic and relational spaces. In this sense, digitalization should not be reduced to a merely technical resource; rather, it should be assessed in relation to accessibility, agency and the institutional conditions that make participation possible.

Comparatively, European initiatives such as INCLUDEED (Erasmus+, 2019–2022) and the German adult integration language provision via the Volkshochschulen point to the value of participatory, blended models and accessibility-by-design for migrant language and civic inclusion. Relative to these examples, the respondents in this study express high expectations regarding digital tools but more cautious evaluations regarding accessibility and alignment with learners' needs, suggesting an implementation gap rather than a consolidated inclusive model.

The chromatic distribution highlights a broadly positive convergence around the perceived usefulness of digital tools for language learning and participation-related dimensions. At the same time, more critical areas emerge with regard to the accessibility of platforms for vulnerable students and the coherence between digital strategies and the actual needs of learners. These patterns reinforce the need for a systemic rethinking of university action grounded in co-design and epistemic inclusion (Fricker 2007; Medina 2013), while remaining within the limits of perception-based evidence.

These patterns point to a structural gap between the perceived potential of digital inclusion and its actual institutional implementation. This gap highlights the need for a shift from technological adoption to pedagogically grounded digital integration.

Greater attention should also be devoted to the direct involvement of migrants in the design and evaluation of digital language-literacy programs. As suggested by Pimmer et al. (2023), migrants' agency should be considered an epistemic and planning resource. From this perspective, opening spaces for co-construction, where migrant students can contribute to the design of content, the evaluation of materials, and the definition of accessibility criteria, appears less as a proven outcome of the present study than as a coherent implication of the theoretical and policy framework adopted here.

Accordingly, the transformative potential attributed to digital technologies should be understood as a normative and policy-oriented perspective rather than as an empirically demonstrated outcome of this study.

This study therefore contributes less to measuring impact than to clarifying how digital inclusion is conceptually framed and institutionally interpreted within contemporary higher education.

5.1 | Key Contributions of the Study

This study offers three main contributions. First, it provides an exploratory map of how university stakeholders and educators in Italy perceive the role of digitally supported Italian L2 literacy in migrant inclusion, with Figure 12 offering a synthetic visualization of the main digital-inclusion block. Second, it identifies concrete perceived gaps, especially around accessibility and the alignment of strategies with learners' needs (Figure 12), that can inform training and institutional planning. Third, it situates these perceptions within the broader policy frame of Twin Transitions and the concept of epistemic inclusion, while also clarifying the limits of a perception-based design.

This contribution remains rooted in the tradition of inclusive pedagogy and in care-based approaches (Mortari 2023a, 2023b).

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In this sense, the university can be interpreted as a public learning space that should foster horizontal, vertical and transversal collaboration capable of recognizing migrants not only as beneficiaries of educational action, but also as participants in knowledge-production processes.

Since the present study captures perceptions rather than measured learning or inclusion outcomes, the evaluation framework outlined below should be understood as a conceptual implication and a proposal for future validation rather than as a direct empirical result.

As a next step, beyond the scope of the present survey, it would be useful to develop an integrated evaluation framework for digitally supported Italian L2 pathways for migrants and refugees across dimensions such as language effectiveness, digital accessibility, educational well-being, civic participation and educational agency. Such a framework would require operationalization and validation through outcome-oriented and comparative studies.

6 | Conclusions

The findings indicate that universities are widely perceived as potentially relevant spaces for the linguistic and digital inclusion of migrants through the integration of educational technologies into Italian L2 teaching. The post hoc robustness checks reported above support reading the main digital-inclusion items as a broadly coherent perception-based block, but they do not convert the study into an outcome evaluation. The targeted use of digital tools therefore emerges not as a demonstrated solution, but as a pedagogical possibility associated by respondents with accessibility, participation and educational support within the broader values of the European Higher Education Area (European Commission 2020).

From this perspective, universities are called not only to respond to emerging educational needs, but also to reflect on their public mission in relation to social cohesion and inclusive educational design (Kauppinen and Kaidesoja 2014). Educational outreach to vulnerable students, including migrants and refugees, is consistent with the strategic objectives of Erasmus+2021–2027 and the European Strategy for Universities (European Commission 2022), which foreground linguistic accessibility, cultural diversity, digitalization and active citizenship.

In practical terms, the study suggests the importance of overcoming fragmented approaches to linguistic inclusion and of fostering stronger integration between educational planning, digital innovation and Third Mission initiatives. University institutions may therefore benefit from strengthening partnerships with local authorities, third sector organizations and European networks in order to support accessible and more personalized learning paths.

The study also offers a basis for future comparative and transnational research capable of examining how different European academic systems address inclusive intervention models. Projects under Erasmus+ Key Action 3 or Horizon Europe, particularly in the areas of democracy and governance, may be strategic for consolidating a shared framework for language education for inclusion grounded in stronger empirical evidence (OECD 2023; UNESCO 2022).

In this sense, promoting accessible and participatory digital language literacy may represent not only an ethical and educational priority, but also an important field in which the democratic function of universities can be further tested and developed within the context of Europe's Twin Transition.

Considering the findings and the transformations affecting higher education in the context of Twin Transitions, some operational recommendations may be advanced for the key stakeholders involved in designing and implementing inclusive education policies. These recommendations concern universities, policymakers and the third sector, and should be understood as policy-oriented implications consistent with the perception-based evidence discussed above.

For universities, a priority lies in strengthening an integrated and systemic approach to linguistic and digital inclusion, through coordinated teaching, technological services and Third Mission activities.

For policymakers, the key implication is to support inclusive innovation by embedding 'inclusion by design' criteria within educational policies and funding frameworks.

The third sector plays a crucial role in fostering local partnerships and co-design processes that enhance accessible and participatory digital language learning pathways for migrant learners. The effective implementation of these recommendations requires a governance approach grounded in shared responsibility, participation and social justice.

Ultimately, the study contributes to reframing digital language literacy not as a technical solution, but as a relational and institutional process whose inclusive potential depends on how universities interpret and enact their public mission within the evolving landscape of the Twin Transition.

Author Contributions

Sara Pellegrini: conceptualization, methodology, investigation, writing – original draft, writing – review and editing, software, data curation, validation. **Riccardo Sebastiani:** conceptualization, methodology, investigation, writing – original draft, writing – review and editing, supervision, visualization, formal analysis.

Ethics Statement

This study was conducted in accordance with the ethical principles for research in the social sciences and education, complying with Italian and European data protection regulations (EU General Data Protection Regulation 2016/679, GDPR) and the principles outlined in the Declaration of Helsinki. Given the nature of the research, an anonymous online questionnaire administered to higher education students and faculty, with no clinical or psychological interventions involved, no formal approval from an Institutional Review Board (IRB) or Ethics Committee was required, as per current guidelines for non-invasive

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educational research. All participants were provided with clear and transparent information about the aims of the study, the voluntary nature of participation, the right to withdraw at any time and the anonymous and aggregated processing of their data. Consent was implied through the voluntary completion of the questionnaire. The collected data were used exclusively for academic research purposes.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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